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INTERCEPTA – MASTER PLAN (end-to-end, first-principles -> fullest vision, with A->Z contingencies)

*The single connected map: where the vision starts, where it ends, every stage between, what to do / what NOT to do / what’s next at each – and for every critical objective a contingency chain (Plan A->B->C?) each triggered by a **pre-registered failure signal**, so we always know WHICH thing is failing and WHY before moving to the fallback. Grounded only in committed evidence (LEDGER.md, experiments/, the Phase-1 audit, the contribution audit). Honest tiers throughout: BUILT - PARTIAL - CLOSED(negative) - OPEN(unsolved science) - GPU-GATED - DATA-GATED - EXPERIMENT-GATED. No vision inflation; gated/open frontiers are named, not pretended solved. Last updated 2026-08-08.**

0. How to read this

- ?1 end state + honest current position. ?2 the 7-stage end-to-end pipeline (the spine). ?3 the ordered master sequence (phases). ?4 the A->Z contingency tree (the heart of the ask). ?5 the closed-door register (what NOT to do). ?6 the resource-gate map. ?7 operating discipline (why it’s non-random). ?8 critical path + immediate next.
 - **The rule that makes it non-random:** every objective has a *pre-registered gate*; failing the gate is the *detection signal* that triggers the pre-planned fallback. We do not improvise fallbacks; they are written here in advance.
-

1. The end state, and the honest current position

Fullest vision (end state): a computational system that, from minimal data, reasons to the best therapeutic intervention for **any** disease – including never-seen ones – rigorously and honestly.

Honest current position (from the two audits, evidence-cited): - ~20-25% of the *fullest computational vision* (Phase-1 audit): deep in ONE scientific domain (metabolic-essentiality target-ID) + validation methodology; shallow/off-the-shelf in structure & ML; near-empty in the molecule/intervention half, systems-biology breadth, simulation/dynamics, foundation-model reasoning, causal modeling. - ~3/10 real-world contribution today (contribution audit): a rigorous, reproducible, honest **map of where zero-data target-ID works and fails**, well-controlled **negatives**, and a fail-safe abstaining engine – no lab-validated target, no drug, no capability the field lacks. - **Two phases (do not mix):** Phase 1 = the strongest computational platform (no wet-lab). Phase 2 = real-world validation (wet-lab, collaborators, clinical). We are in Phase 1.

2. The end-to-end pipeline (the connected spine – 7 stages)

Each stage: what it is - status - the transfer-condition (when its signal is trustworthy) - what feeds it / what it feeds.

S0 – INPUT & representation. Accept a disease as a genome / causal gene / phenotype / screen. - Status: **PARTIAL**. genome ? (pathogens), causal gene ? (MENDEL1), GWAS ? (GENETICS1), phenotype->gene ? retrieval (PHENO1), dependency screen ? (DEPEND1). Missing: raw phenotype->de-novo mechanism (OPEN); imaging/EHR/multi-omics input. - Feeds -> S1/S2 via the autonomous class detector (ROUTERAUTO1).

S1 – MECHANISM understanding. Infer the biology that makes a node matter. - Status: **metabolic BUILT** (FBA, validated VAL-ESS/CROSSVAL/BLIND1-7); **non-metabolic DEFINITELY CLOSED** – 4 signal classes each failed (MET4 PPI, NONMET1 synteny, REGNET1 regulatory, PLMESS1 PLM-embedding) AND their ensemble upper-bound failed (MULTISIG1); conservation-breadth (0.908) is the accepted ceiling. **regulatory/signaling/dynamic/whole-cell = MISSING/OPEN**. - Transfer-condition: FBA transfers where the GEM encodes genuine biosynthetic dependence (fails on host-scavengers, sparse GEMs; META1/FAIRGATE bound it).

S2 – TARGET identification. Which protein/node to intervene on. - Status: **BUILT (narrow-but-validated)**. essentiality (FBA), conservation-breadth (the non-metabolic workhorse, AUROC 0.91), structural class-ID (viruses, GENERALIZE2/3), functional dependency (cancer, DEPEND1), genetics (complex, GENETICS1), causal-gene (monogenic, MENDEL1). Silent-failure on phylogenetically-isolated organisms (TID4) is a known limit.

S3 – PRIORITIZATION, SAFETY, EXPERIMENT-DESIGN. Rank, filter unsafe, decide what to test. - Status: **BUILT**. multi-axis best-intervention (BESTINT1), hard host-non-homology safety filter (FRONT1/E2E2), resistance/condition-robustness, calibrated confidence (CALIB1), VOI experiment-prioritization (EXPDESIGN1, ~8x wet-lab efficiency). Honest bound: durability from WHOLE-PROTEIN static biology = CLOSED (AMR1) – but drug-CONTACT-residue DYNAMICS gives a first signal (DYNAMICS1, AUROC 0.84, n-fragile); OOD abstention guarantee degrades (CONFORMAL1: 94%->55%).

S4 – INTERVENTION design. Target -> therapy. - Status: **PARTIAL/mostly-unbuilt (the weak half)**. modality triage ? (MODALITY1, fail-safe), repurposing ?-narrow (INTERVENE1/2, ~1-7% coverage), ADMET ? (B30), synthesizability ? (B31), generation ?-standalone (B33). **Novel-target binding affinity** – standard tools FAIL (docking~chance HIT2; QSAR analog-bound HIT1; PCM null B49; active-learning null B65); co-folding (Boltz-2) is the credible SOTA but is **GPU-gated even at small-candidate scale** (empirically confirmed 2026-08-09: a CPU pilot on ~20 candidates produced 0/20 in ~48 min and was stopped – the affinity head’s diffusion sampling is the expensive part; an earlier “CPU-feasible small-candidate” reframe was tested and RETRACTED). Structural repurposing = CLOSED (promiscuity, STRUCTREPURPOSE1).

S5 – VALIDATION. In-silico confidence -> wet-lab -> clinical. - Status: in-silico ? (reproduced x2, blind protocol); **wet-lab = EXPERIMENT-GATED** (CRISPRIDESIGN1 turnkey design ready, needs a lab); **clinical = EXPERIMENT-GATED + tested-negative** for response prediction (B10/B20).

S6 – INTEGRATION & GOVERNANCE. Route, abstain, calibrate, reproduce. - Status: **BUILT.** composite router + transfer-condition gating + abstention (COMPOSITE1-4), autonomous class detection (ROUTERAUTO1), end-to-end integration proof (CAPSTONE2, fail-safe), verdict-stable reproducibility. The program’s genuine strength alongside methodology.

3. The master sequence (ordered by leverage; what-to-do / what-NOT / next)

WAVE 1 – CPU-feasible Phase-1 inventions (do now, no new resources). Ordered by leverage: 1. PLMESS1 (DONE -> 4th closure) + MULTISIG1 (DONE -> the ensemble-ceiling closure). The non-metabolic-mechanism question is now **definitively closed** (individually and combined); conservation-breadth (0.908) is the accepted ceiling. 2. Uncertainty productionization – ship class-conditional (Mondrian) conformal into the engine (CONFORMAL1 showed marginal is vacuous for targets); genuine, honest, closes a real hole (NOTE: CONFORMAL1 also showed it degrades OOD – productionize with the OOD-widening rule, ?19). 3. Knowledge-integration – a calibrated multi-signal posterior with honest joint uncertainty (careful: MULTISIG1 already showed the non-metabolic signal-union is collinear with conservation, so gains here are modest by construction). - **What NOT to do in Wave 1:** any further non-metabolic-mechanism attempt – CLOSED individually (4 signals) AND at the ensemble upper-bound (MULTISIG1); more shallow disease-class coverage arms (diminishing-returns); anything requiring GPU dressed up as CPU.

WAVE 2 – GPU-gated Phase-1 (the biggest % of the remaining vision; needs a GPU – a resource decision). In priority: 1. Zero-shot affinity via co-folding (Boltz-2) – the 14-weight intervention half; spec ready (AFFINITY1 GPU_BENCHMARK_SPEC). 2. Foundation-model mechanism reasoning for the non-metabolic half (only if PLMESS1/embeddings hint signal). 3. MD / dynamics (resistance evolvability – the AMR1 follow-on; PK/PD).

WAVE 3 – DATA-gated Phase-1. Curated eukaryote GEMs (lift the blind-eukaryote fails); a genome-wide *P. falciparum* CRISPR screen (resolves the parasite confound); larger curated regulatory/interaction networks (only if a fresh signal-class warrants).

PHASE 2 – real-world (only after Phase-1 is genuinely mature; per the sequencing directive). Wet-lab CRISPRi of a nominated target (CRISPRIDESIGN1) -> collaborator; then prospective novel-pathogen test; then clinical.

PARALLEL CHANNEL – dissemination. Publish the *validated core + negative map* (the genuine ~3/10 contribution) at a methods/reproducibility venue; it does not require Phase-1 completion and delivers the real value to the world. Framing fixes from the red-team (ANTICIPATED_REVIEWS.md) applied first (soften “prospective-blind”/“law”; wall off post-hoc).

4. THE CONTINGENCY TREE – Plan A->Z per critical objective, each with its FAILURE-DETECTION signal

Format: A primary -> [detection: the pre-registered signal that proves A failed] -> B fallback -> ? -> terminal honest fallback* (accept the bound / route to experiment / abstain). We move A->B without pause, but only after the detection signal fires – never on a hunch.*

O1 – Accept ANY disease as input

- **A:** genome/gene/GWAS/screen/phenotype inputs via the autonomous detector. [*detection: detector mis-routes or over-fires on a class – caught by the CAPSTONE2 fail-safe test / 93 unit tests.*]
- **B:** if a new input type has no validated route -> **ABSTAIN with an explicit “unsupported input” flag** (never guess).
- **Terminal:** raw-phenotype->de-novo-mechanism is **OPEN** – we honestly restrict to retrieval over known maps (PHENO1), not de-novo inference. Do NOT fake de-novo inference.

O2 – Infer NON-METABOLIC mechanism (the #1 core gap)

- **A:** FBA metabolic essentiality. [*status: BUILT + validated for the metabolic half.*]

- **B:** PPI-network centrality (MET4). *[detection: lift collapses under a study-bias control (+0.128->?0.004) -> FAILED.]*
- **C:** conserved genomic context / synteny (NONMET1). *[detection: DeltaAUROC-beyond-conservation +0.016 < +0.03 gate -> FAILED.]*
- **D:** curated regulatory master-regulator influence (REGNET1). *[detection: OR 0.52, DeltaAUROC ?0.006 -> FAILED, clean null.]*
- **E:** learned PLM (ESM-2) embeddings (PLMESS1). *[detection: DeltaAUROC-beyond-conservation +0.008 < +0.03; study-bias-ctrl ?0.0006 -> **FAILED** 2026-08-08, reproduced, baseline consistency-checked vs NONMET1.]*
- **F (the ENSEMBLE upper-bound, not a new signal):** combine all four signals (MULTISIG1). *[detection: full-ensemble DeltaAUROC-beyond-conservation +0.019 logistic / +0.009 GBM < +0.03; drop-one ablation - no signal clears; embedding not the driver -> **FAILED** 2026-08-08, reproduced (sha e6badcb7).]* This is the honest way to close the question definitively without a random 5th *new signal*: test the best-case COMBINATION.
- **Terminal honest fallback – NOW ACTIVE & DEFINITIVE (E and F both failed):** the non-metabolic mechanism is closed **both individually (4 signal classes) AND at the ensemble upper-bound (MULTISIG1)** – a decisive scientific bound. **Conservation breadth (AUROC 0.908) is the accepted, unbeatable ceiling** for the non-metabolic half. The **ONLY** untested lever is a larger/structure-aware PLM or mechanism data (GPU/DATA-gated, Wave-2/3). **No further CPU attempts** – the objective is completed and closed, not open.

O3 – Identify a TARGET for a novel organism/disease

- **A:** the class-matched validated signal (FBA/dependency/genetics/structure) at earned confidence. *[detection: the signal’s pre-registered gate (OR>3/RR-fair, etc.) fails on the class.]*
- **B:** if the primary signal is out-of-domain -> fire the supporting signal (conservation-breadth + host-safety).
- **C:** if none transfers (dark proteome / novel zero-screen parasite) -> **ABSTAIN** (DARK1/TRANSFER1 proven fail-safe).
- **Terminal:** abstention is a valid answer – “no trustworthy target from computation alone; needs experimental data.”

O4 – Prioritize + guarantee SAFETY + spend experiments well

- **A:** hard host-non-homology safety filter + multi-axis score + VOI experiment-prioritization (EXPDESIGN1). *[detection: a host-toxic target reaches the shortlist – caught by the hard filter’s construction test.]*
- **B (durability):** predict resistance-liability. **A** = whole-protein static biology (AMR1) *[detection: AUROC 0.556 -> FAILED]* -> **B** = drug-CONTACT-residue DYNAMICS (DYNAMICS1, ESM entropy at contact residues) *[AUROC 0.839, MWU p 0.029, beats AMR1; **PARTIAL PASS** 2026-08-08 – but n-FRAGILE (n=15; subset p 0.05-0.12), a demonstration not a validated predictor]* -> **C** = larger target set / fitness-scan / experimental confirmation (DATA/EXPERIMENT-gated) to firm up n.
- **B (abstention trust):** conformal confidence. *[detection: CONFORMAL1 – OOD essential-class coverage 55% « 90%.]* -> fallback: **make OOD abstention more conservative** / report confidence as not-guaranteed-out-of-distribution (do not claim nominal coverage on a novel organism).

O5 – Choose the intervention MODALITY

- **A:** mechanismxlocalization fail-safe recommender (MODALITY1). *[detection: an infeasible modality is recommended – caught by the hard fail-safe (0/43).]*
- **B:** where features are absent or the class is out of MODALITY1’s validated (human-disease) scope -> **ABSTAIN** (pathogen modality currently abstains – honest).
- **Terminal:** modality is a CLASS recommendation, never a molecule; the molecule is O6.

O6 – Produce a real INTERVENTION for the target (the hardest objective)

- **A:** repurpose an existing drug (INTERVENE1/2). *[detection: novel-target coverage ceiling – ~1/32; measured.]*
- **B:** structural repurposing to expand coverage (STRUCTREPURPOSE1). *[detection: gain is a promiscuity artifact under a null -> FAILED.]*
- **C:** zero-shot binding-affinity ranking for a novel target – docking (HIT2), QSAR (HIT1), PCM (B49), active-learning (B65). *[detection: each ~ chance / null on the pre-registered gate -> all FAILED on CPU.]*
- **D: co-folding (Boltz-2) affinity – GPU-GATED even at small-candidate scale (EMPIRICALLY CONFIRMED 2026-08-09).** A prior deep-analysis note claimed this was CPU-feasible for ~20-candidate ranking (~3 h); I RAN it and that claim was WRONG – the CPU pilot produced **0/20 predictions in ~48 min wall** / **~90 min CPU** and was stopped as impractical. Root cause: my ~10 min/complex figure was AFFINITY1’s *structure-only* time; the **affinity head adds many diffusion samples**, pushing per-complex CPU time to tens of

minutes-hours -> even ~20 complexes don't finish in a session. So the reframe is **RETRACTED**: affinity co-folding needs a GPU. *[detection stays the GPU benchmark: AUROC ≥ 0.60 AND > 0.4285 AND novel-vs-inactive ≥ 0.60 . See experiments/AFFINITY1_cofolding_zeroshot/CPU_PILOT_FINDING.md.]*

- **E**: physics-based (MM-GBSA / FEP-lite) rescoring – GPU/compute-gated.
- **Terminal honest fallback**: if D/E fail or stay unavailable -> **novel-target affinity is OPEN science**; route the validated target to **experimental screening** (Phase 2) rather than emit a fake potency-ranked lead. NEVER present a pose-plausible hypothesis as a validated lead.

O7 – VALIDATE in reality (cross from Phase 1 to Phase 2)

- **A**: wet-lab CRISPRi knockdown of a nominated target (CRISPRIDESIGN1 turnkey). *[detection: growth defect $\geq 5x$ vs control -> confirmed; $< 5x$ -> the prediction is WRONG (first-class negative) -> recalibrate the signal that nominated it.]*
- **B**: if no lab/collaborator -> the collaboration ask + turnkey protocol + the preprint are the instruments to GET one (a professor, a course, a neglected-disease effort). *[detection: no partner after outreach -> escalate channels.]*
- **C**: cheaper variant – MIC of a commercial pathway inhibitor vs the organism (if a drugged target).
- **Terminal**: until a wet-lab result exists, all target/molecule outputs remain **pre-registered hypotheses** – stated as such, never as validated.

O8 – INTEGRATE + govern “any disease”

- **A**: composite router fires validated signals per class, abstains otherwise (CAPSTONE2). *[detection: a mis-fire or a verdict-skeleton drift – caught by the pre-registered CAPSTONE2 gate + regression tests.]*
- **B**: new validated arm -> add additively (verdict-stable); unvalidated -> abstain. *Do NOT expand coverage without a passing validation.*

O9 – DISSEMINATE (deliver the real contribution)

- **A**: publish the validated core + negative map (methods/repro venue) + preprint. *[detection: reviewer novelty objection – pre-empted by ANTICIPATED_REVIEWS.md; framing fixes applied first.]*
- **B**: if the flagship is judged insufficiently novel -> reframe as a negative-results / reproducibility paper (its genuine strength) rather than a discovery paper. *Do NOT inflate to force a discovery framing.*
- **C**: collaboration outreach in parallel (the wet-lab ask is the lever above 3/10).

5. The CLOSED-DOOR / ANTI-PATTERN register (what NOT to do – with evidence)

1. **Do NOT re-attack the non-metabolic mechanism** – CLOSED individually across 4 signal classes (MET4/NONMET1/REGNET1/ PLMESS1) AND at the ensemble upper-bound (MULTISIG1) – accept the conservation ceiling (0.908); the next move there is GPU (structure-aware PLM) or experimental data, not another CPU signal or combination.
 2. **Do NOT use docking/QSAR/PCM to claim novel-target potency** (HIT2/HIT1/B49/B65 negatives).
 3. **Do NOT predict patient drug-RESPONSE from baseline profiles** (B10/B20 tested-negative).
 4. **Do NOT rank targets by conservation as if safe** (FRONT1: most-conserved = host-toxic); use the hard host-non-homology filter.
 5. **Do NOT use structural repurposing for coverage gain** (STRUCTREPURPOSE1 promiscuity).
 6. **Do NOT predict resistance-liability from WHOLE-PROTEIN static biology** (AMR1 null) – use drug-CONTACT-residue DYNAMICS (DYNAMICS1: AUROC 0.84, beats AMR1; n-fragile, treat as a demonstration pending larger-n/experimental confirmation).
 7. **Do NOT trust a marginal/OOD abstention guarantee at nominal rate on a novel organism** (CONFORMAL1: 94%->55%).
 8. **Do NOT spawn shallow coverage arms or run GPU-work on CPU or overclaim** (audits).
 9. **Do NOT do random trial-and-error** – every step pre-registered with a failure signal (this document).
-

6. Resource-gate map (so we always know WHY something is blocked)

- **CPU-now (do autonomously)**: Wave-1 items; the parallel publication prep.

- **GPU-gated (resource decision – the single highest-leverage unlock, the 14-weight intervention half):** co-folding affinity (AFFINITY1 spec), foundation-model reasoning, MD/dynamics.
- **DATA-gated:** curated eukaryote GEMs; a Pf CRISPR screen; larger curated networks. (Fetchable open data is NOT gated – proven by autonomously sourcing the Abasy GRN for REGNET1.)
- **EXPERIMENT-gated (Phase 2, needs a person + a bench):** wet-lab validation, clinical.

7. Operating discipline (why this plan is non-random)

Pre-register every gate before running - reproduce x2 byte-identical - control nulls/leakage/study-bias - report negatives first-class - honest tiers - abstention-integrity over coverage - **each objective's failure-detection signal is written in advance (?4), and the fallback is pre-planned – we never improvise the next move.** A negative is a completed step (it closes a door and redirects resource), not a failure to hide.

8. Critical path & immediate next actions

- **NOW (CPU, autonomous):** the non-metabolic mechanism line is DEFINITELY CLOSED (PLMESS1 4th closure + MULTISIG1 ensemble-ceiling closure, both verified/reproduced). Remaining CPU-buildable = modest ENGINEERING only (uncertainty productionization, knowledge-graph integration); the high-value CPU frontier is exhausted.
 - **THE ONE RESOURCE THAT MOVES THE % MOST:** a GPU -> unlocks the intervention half (O6-D) – the biggest single chunk of the remaining ~75-80%. Name it; do not fake around it.
 - **THE ONE THING THAT MOVES REAL-WORLD CONTRIBUTION MOST (Phase 2):** one **wet-lab CRISPRi test** (O7-A) – converts the whole in-silico edifice to confirmed; needs a collaborator (O7-B / O9-C).
 - **PARALLEL:** publish the validated core + negatives (O9-A) – delivers the genuine contribution now.
-

9. Definition of done + honest completion

- **Phase-1 computational done** when: every reachable disease class is covered-or-honestly-abstaining; both halves are built-or-honestly-bounded; integration is proven end-to-end; every wall is characterized; and the GPU/data-gated items are either done (given resources) or precisely specced. **Current: ~20-25% of the fullest computational vision** (audit).
 - **Phase-2 real-world done** when: ≥ 1 nominated target is wet-lab-confirmed; a collaborator reproduces; and – the far end – a validated intervention. **Current: 0% (no wet-lab).**
 - The honest end state we can reach on CPU + open data alone is a **rigorous, validated, honest ~one-fifth computational platform + a decisive negative map** – everything beyond needs GPU, data, or a bench. This plan names exactly which, and what to do when each is unblocked or each Plan-A fails.
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PART II – the deeper connective layers (dependency DAG - invention/engineering/acquisition - combination matrix - state machine)

10. The dependency DAG (what unlocks what – the true connected critical path)

Objectives/builds as nodes; “->” = enables/unlocks. This is the connective tissue: it shows the LONGEST POLES, not just a list.

```

S0 input+autodetect --> S2 target-ID -> S3 prioritize/safety -> S5(in-silico confidence) --+
(BUILT)          |          (BUILT)          (BUILT)                                     |
S1 mechanism -----+                                                                |
+ metabolic FBA (BUILT) -----> feeds S2      |
+ non-metabolic (CLOSED: 4 signals + ensemble MULTISIG1) --X (conservation-breadth is the ceiling) |
                                                                v
S4 intervention:  modality (BUILT) -> repurpose (narrow) --X coverage ceiling          S6 governance
                                                                +> NOVEL-TARGET AFFINITY ==[GPU-GATE]==> co-folding (BUILT: router,

```

S7 VALIDATION: in-silico -----==[WET-LAB GATE]==> CRISPRi test -> prospective -> clin
 09 DISSEMINATION: publish validated core+negatives (PARALLEL -- blocks nothing, delivers the real ~3/10 now)

Longest poles (the two gates that gate everything downstream): 1. **GPU gate** -> unlocks novel-target affinity (O6-D) -> the entire **intervention half** (14 weight) -> real molecule outputs -> meaningful molecule-side Phase-2. *Nothing on the molecule side moves without it.* 2. **Wet-lab gate** -> unlocks O7 -> converts the whole in-silico edifice to **confirmed** -> the ~3/10->higher real-world jump -> prospective -> clinical. *Nothing real-world moves without it.* **Everything else is either already BUILT (left of the gates) or a modest CPU deepening that does not cross a gate.** So the connected truth: **two acquisitions (a GPU, a bench) are the critical path to the fullest vision – not more CPU experiments.**

11. Invention vs Engineering vs Acquisition (honest tagging – “every step an invention” is NOT literally true)

The directive says “every step a genuine invention.” Truth-over-vision requires I disagree precisely: the remaining steps split three ways, and the biggest %-movers are ACQUISITIONS, not inventions. Pretending otherwise would be a false claim.

- **GENUINE INVENTIONS still required (open science – nobody has solved these; do NOT fake):** (a) zero-shot novel-target binding-affinity ranking (O6 – the field’s open wall); (b) de-novo mechanism inference from a raw phenotype (O1 deep case); (c) causal disease modeling. (Resistance-evolvability dynamics – formerly on this list – is now **PARTIALLY BUILT**: DYNAMICS1 gave a first contact-residue durability signal, AUROC 0.84 n-fragile; firming it up is DATA/EXPERIMENT-gated, not a new CPU invention.) These are where “invention/novelty” genuinely lives – and most are GPU- or data-gated, or genuinely unsolved.
- **CLOSED – no longer an open invention target:** a homology-independent non-metabolic MECHANISM signal on CPU is DEFINITELY closed (4 signal classes + the MULTISIG1 ensemble upper-bound all fail to beat conservation-breadth 0.908); the only untested lever is a GPU-scale structure-aware PLM or experimental mechanism data – an ACQUISITION, not a CPU invention.
- **ENGINEERING (known method, just build/run – NOT inventions; call them what they are):** class-conditional-conformal productionization; knowledge-graph integration; autonomous-detector refinement; modality-recommender generalization; **running** co-folding/MD (the methods exist – the block is GPU compute, not invention).
- **ACQUISITION (get a resource/data – zero invention, highest leverage):** a GPU; a wet-lab collaborator; curated eukaryote GEMs; a genome-wide *P. falciparum* CRISPR screen; larger curated networks. **Honest breakdown of the remaining ~75-80%:** it is dominated by **acquisitions + a few genuinely-open inventions**, with a thin layer of engineering. This is why “keep spawning CPU experiments” cannot close it – and why the plan’s top actions are acquisitions + publication, not an infinite CPU experiment stream.

12. The COMBINATION / SCENARIO matrix (the explicit “other combination plans” – parallel & cross-objective)

*Beyond the per-objective A->Z chains (?4), the real world presents COMBINATIONS of outcomes. Here is the pre-planned response to each combination, so we “go without pause” but never randomly. Rule: **publication (O9) + acquisition outreach (GPU + collaborator) run in PARALLEL in every scenario** – they block nothing and deliver/unlock the most.*

Scenario (combination of states)	Detection	Pre-planned combined response
S-A: PLMESS1 fails + no GPU + no bench (most-likely near-term)	PLMESS1 DeltaAUROC<0.03/leakage; no resources	Consolidate + publish the validated core+negatives (O9); pursue GPU + collaborator acquisitions in parallel; STOP CPU experiment spawning (compute frontier mapped). Honest default.

Scenario (combination of states)	Detection	Pre-planned combined response
S-B: PLMESS1 succeeds	DeltaAUROC $\geq$ 0.03, survives leakage+study-bias	Bank the foundation-model non-metabolic signal; extend to more organisms (data-permitting, ENGINEERING); integrate into the engine; update manuscript. Then S-A for the rest.
S-C: GPU acquired	resource available	Run AFFINITY1 GPU benchmark (O6-D). Pass->intervention half opens (biggest %-jump); Fail->physics/MD (O6-E); Fail->route target to experimental screening (O6-terminal). In parallel: foundation-model mechanism (O2-E'/causal).
S-D: wet-lab collaborator secured	partner + bench	Run CRISPRi (O7-A) on the top broad-spectrum nominated target. $\geq$ 5x defect-> first real-world confirmation (the ~3/10->higher jump); $<$ 5x->prediction wrong->recalibrate the nominating signal (first-class negative).
S-E: GPU AND bench both absent long-term	time passes, no resources	Terminal honest state: the vision is bounded at the validated compute-only platform + negative map until resources arrive; publish, keep the ask open, do not fake progress . This is an acceptable, honest end for the autonomous phase.
S-F: publication rejected as “insufficient novelty”	editor/reviewer verdict	Reframe as a negative-results/reproducibility paper (its genuine strength, per ANTICIPATED_REVIEWS); bioRxiv preprint stands regardless; the contribution is unchanged.
S-G: a committed result fails to reproduce / a confound found	reproduce-x2 mismatch / null control	Retract/down-tier in the LEDGER immediately (as done for V14/N1, the “host-embedded” rule, EXPDESIGN1-G2, CONFORMAL1 marginal); integrity over preservation.

Combined-fallback logic (no pause, not random): at any moment, execute the S-row matching the current state; when a detection signal fires, transition to the matching S-row; keep O9 + acquisition-outreach running in parallel throughout.

13. Current-state decision table (given the state RIGHT NOW -> exact next action)

- **State now (updated 2026-08-08):** PLMESS1 **DONE** -> 4th closure AND MULTISIG1 **DONE** -> **ensemble-ceiling closure** – the non-metabolic mechanism is now closed **individually (4 signals) AND at the ensemble upper-bound**; **Scenario S-A is ACTIVE**; no GPU; no wet-lab; Phase-1 ~20-25% (of the fullest COMPUTATIONAL platform; ~5-8% of the fullest END-TO-END vision, Part IV); MASTER_PLAN I-IV + contingency tree in place; publication honesty-pass + manuscript Part-III fold-in + collaboration-brief refresh done.
- **Immediate next (autonomous, per S-A):** the non-metabolic CPU line is **DEFINITELY CLOSED** (individual + ensemble – no further attempts). The remaining genuinely-CPU-buildable items are modest (uncertainty production-ization, knowledge-graph integration) and are honestly ENGINEERING, not needle-movers. The CPU-experiment frontier is at its honest end – the needle-movers now require acquisitions (GPU -> intervention half; a bench -> real-world + the closed loop) that need the human.
- **Parallel-available-now (CPU, autonomous, modest):** remaining publication prep – a $\leq$ 25-word title; the off-class-TM and study-matched-cancer nulls (small re-analyses); these are ENGINEERING finishing touches, not

needle-movers.

- **Blocked-pending-acquisition (the needle-movers – need the human):** GPU (-> intervention half), wet-lab collaborator (-> real-world jump). **These are the honest top of the backlog and I cannot self-acquire them.**
- **Honest statement of the autonomous frontier:** the CPU-only, open-data, zero-budget invention space is **at its honest end** for the high-value questions. With the non-metabolic mechanism now closed both individually and at the ensemble upper-bound (MULTISIG1), the genuinely value-adding moves are **acquisitions** (GPU, bench) and **dissemination** – not more autonomous CPU experiments. A handful of modest ENGINEERING items remain (uncertainty productionization, knowledge-graph integration); I will label them as such and not present them as needle-movers.

14. Executive synthesis (one screen – nothing missed)

The vision = best intervention for any disease, computationally. We are ~20-25% of the fullest *computational* platform and ~3/10 real-world: **deep in metabolic target-ID + validation methodology, empty in the molecule half / systems-biology breadth / simulation / foundation-model reasoning / causal.** The pipeline (S0->S6) is BUILT through target-ID + prioritization + governance; the **two gates** that gate everything downstream are a **GPU** (-> the intervention/affinity half) and a **wet-lab bench** (-> real-world confirmation). The non-metabolic-mechanism door is now DEFINITELY closed – 4 signal classes individually (MET4/NONMET1/REGNET1/PLMESS1) AND the ensemble upper-bound (MULTISIG1). Every objective has a pre-registered failure signal and a pre-planned fallback (?4), and every *combination* of outcomes has a pre-planned response (?12). What NOT to do is enumerated (?5). The honest top of the backlog is **acquisitions (GPU, collaborator) + publication of the validated core+negatives** – not more CPU experiments, which are near their honest end. Truth over vision: we are 100% ready to *execute* what is buildable/known, honest that the rest needs resources or genuine open-science inventions we name explicitly and do not fake.

PART III – the ULTIMATE connective layer (every dot connected: data-flow contracts - full traceability of 160 experiments - the closed feedback loop - the composition law - cross-cutting threads - failure-cascade - the logical spine)

This part closes the gaps Parts I-II left implicit: it makes every logical connection EXPLICIT – how each stage’s output becomes the next stage’s input, how all 160 committed experiments map onto the pipeline, how results FEED BACK to improve the engine, how per-class models COMPOSE into “any disease,” how uncertainty/safety/bias thread THROUGH every stage, and how a failure at any node CASCADES. Every connection is grounded in a committed experiment; none is invented.

15. The pipeline as a connected data-flow graph (stage I/O contracts – no orphan dots)

Each stage: INPUT => OUTPUT => CONSUMER. Read top-to-bottom = the forward pass; ?17 adds the backward (feedback) edges.

- **S0 Input+detect:** {genome | causal-gene | GWAS | CRISPR-screen | HPO-phenotype} => {biology_class + data-descriptors} => **S1/S2** (via ROUTERAUTO1/class_detector). *Orphan closed:* raw-phenotype enters only via PHENO1 retrieval -> candidate gene -> re-enters as a “causal-gene” input; de-novo-from-phenotype is OPEN and routes to ABSTAIN.
- **S1 Mechanism:** {genome, GEM} => {FBA-essential set + confidence} (metabolic); the non-metabolic branch emits **only conservation-breadth** (4 closures MET4/NONMET1/REGNET1/PLMESS1 + the MULTISIG1 ensemble ceiling) => **S2**. *Connection:* S1’s transfer-condition (GEM encodes real biosynthetic dependence; META1/FAIRGATE) decides whether its output is trusted or capped.
- **S2 Target-ID:** {class, S1 outputs, proteome, reference targets} => {ranked targets + per-target signal-provenance + confidence-tier} => **S3**. *Connection:* the class picks the signal (FBA/dependency/genetics/structure) per the transfer table (?18); silent-failure risk (TID4) is carried as a confidence caveat.
- **S3 Prioritize/Safety/ExpDesign:** {S2 targets} => {host-safe, multi-axis-ranked (BESTINT1), VOI-ordered (EXPDESIGN1) shortlist + calibrated confidence (CALIB1) + explicit abstentions} => **S4 and S5** (which targets to test first). *Connection:* the hard host-non-homology filter (FRONT1/E2E2) is a GATE – nothing unsafe passes to S4.

- **S4 Intervention:** {target + mechanism + localization} => {modality-class (MODALITY1) + repurposing-candidate (INTERVENE1/2) OR an explicit "novel-affinity = OPEN/GPU-gated" flag} => **S5**. *Connection:* the SM-inhibitor branch is gated by the affinity wall (HIT2/AFFINITY1) – it emits a *feasibility class*, never a validated potent molecule.
- **S5 Validation:** {shortlist + intervention} => {in-silico confidence (reproduced x2); a turnkey wet-lab design (CRISPRIDESIGN1); and -- once run -- an experimental result} => **S6 and feedback to S1/S2** (?17). *Connection:* this is the Phase-1->Phase-2 boundary; everything above is in-silico hypothesis, everything at/after the experimental result is real-world.
- **S6 Governance (wraps all):** routes (class detector), **abstains** where no signal transfers, propagates confidence, enforces provenance/reproducibility, composes (?18). *Connection:* S6 is the only stage that can emit a *final* answer or a *final abstention* – it is the integrity gate on the whole graph.

16. Full traceability – all 160 experiments mapped onto the pipeline (every dot -> its objective + status)

Grouped by pipeline node (representative IDs; status in caps). This connects the entire experimental corpus to the plan.

- **S2 Target-ID core (metabolic) – VALIDATED:** MET1/MET2/MET3 (mechanism breaks the conservation ceiling), VAL-ESS/CROSSVAL (6 curated GEMs, OR 4-45), PREDVAL (per-target scorecard), REACH1 (conservation-breadth recovers non-metabolic essentials), BESTINT1 (multi-axis). Blind: BLIND1/2/3/6 PASS, BLIND4/5/7 FAIL (analyst-blind, 4/3 split).
- **S1 Mechanism non-metabolic – CLOSED (4 signals + ensemble):** MET4 (PPI/study-bias), NONMET1 (synteny/collinear), REGNET1 (regulatory/clean null), PLMESS1 (PLM/re-encodes conservation), + MULTISIG1 (the ensemble of all four still doesn't beat conservation). => conservation-breadth (0.908) is the accepted, definitive ceiling.
- **S2 Target-ID other classes:** viruses -> FOLD1/2, GENERALIZE1/3, HARDENV1 (structural class-ID, PASS n=5); fungi/eukaryote -> GENERALIZE4, HARDENF1 (PASS); parasites -> GENERALIZE5, HARDENP1, HOSTCTX1/2, PARAREOLVE1/2 (GEM/base-rate-bounded); cancer -> DEPEND1, F3CLIN1 (dependency + patient-driver); monogenic -> MENDEL1; complex -> GENETICS1; phenotype-input -> PHENO1.
- **S2 Target-ID negatives/bounds:** TID1/2 (conservation ceiling), TID3 (kingdom degradation), TID4 (silent failure), TRANSFER1 (dependency doesn't transfer label-free).
- **S3 Prioritize/Safety/ExpDesign:** FRONT1/FRONT2/E2E2 (selectivity/safety), CONDROB1 + SYNLETH1/2 (robustness), CALIB1 (calibration), EXPDESIGN1 (VOI ~8x efficiency), AMR1 (durability – CLOSED for static biology), CONFORMAL1 (OOD abstention bound).
- **S4 Intervention:** MODALITY1 (modality triage), INTERVENE1/2/3 (repurposing, narrow), B30/B30b (ADMET), B31 (synthesizability), B33/B40/B52 (generation). **Affinity wall (CLOSED/GPU-gated):** HIT1/HIT2, C1, B46/B47/B48/B49, B55/B56/B63, B65, AFFINITY1 (co-folding, GPU-gated), STRUCTREPURPOSE1 (promiscuity).
- **S5 Validation + Feedback:** CRISPRIDESIGN1 (turnkey wet-lab design), EXPVAL/BROADSPEC/DRUGGABLE/PANBAC (deployment predictions), NEWBUG/SAUREUS (held-out); **feedback:** SIL1 (self-improving loop works in-domain), SIL2 (washes under shift), B45/B51/B65 (active-learning bounds).
- **S6 Governance/Integration:** SUBSTRATE1-5, ENGINE/engine_v1, discovery_engine, COMPOSITE1-4, ROUTERAUTO1, DARK1 (abstention-integrity), CAPSTONE1/2 (end-to-end proofs); method inventions META1 (transfer principle), FAIRGATE1 (base-rate gate).
- **Era-1 cancer (bounds that shaped the plan):** B1-B23 – drug-response intrinsic ceiling (B1/B2), clinical null (B10/B17), external-replication failure (B20/B21) => the “clinical response prediction = tested-negative” bound; synergy B24-B29 (PARTIAL).

17. The CLOSED FEEDBACK LOOP (the biggest missing connection – forward<->backward)

Parts I-II were a forward pass. The fullest vision is a LOOP: an experimental result must flow BACK to improve the engine. That edge exists in evidence but was never wired into the plan. Here it is, with its honest bound. - **Forward:** S0->?->S5 nominates a target/intervention + a turnkey experiment (CRISPRIDESIGN1). - **Backward (S5 => S1/S2):** a wet-lab result (or any new labeled data) re-enters via the substrate's **continuous-absorption guardrail** (SUBSTRATE3) as high-tier evidence, and **recalibrates the nominating signal** (a confirmed essential raises that signal's weight; a refuted one lowers it). Mechanism proven in-silico: **SIL1** (conformally-gated self-improving loop improves in-domain, 6/6 tasks) – but **SIL2** (the loop washes out under distribution shift) and **CONFORMAL1** (OOD confidence degrades 94%->55%) together impose the HARD RULE: the feedback loop may only trust self-/near-domain labels; on a novel

organism it must widen uncertainty and defer to real experimental labels – never bootstrap confident labels out-of-distribution. This connects the negatives (SIL2, CONFORMAL1) to the loop’s safe design. - **Consequence for the vision:** true closed-loop autonomy is **gated on real experimental labels** (Phase 2). Until then the loop safely improves only within validated domains. This is why O7 (wet-lab) is the master unlock – it is the **ONLY** input that closes the loop honestly.

18. The COMPOSITION LAW (“book of validated models” -> “any disease”)

How the pieces become the whole – the meta-level connection. “Any disease” is NOT one model; it is the **union of per-class validated signal-models, each fired only inside its transfer domain, composed by the router, with abstention everywhere else** (composite-architecture reframe). The **transfer-condition principle IS the composition operator:** `answer(disease) = (+)_class [ signal_class if transfer-condition(class) holds, at earned confidence ] ; else ABSTAIN.` - Coverage today = the union of validated domains: bacteria/archaea (FBA full), fungi (FBA capped), viruses (structure), cancer (dependency), monogenic (mode), complex (genetics capped), + conservation-breadth everywhere as the workhorse. - The frontier = the complement (where no model transfers) -> **honest abstention** (DARK1/TRANSFER1). Growth of “any disease” = adding validated domains (each a new “chapter”), never forcing one model onto biology it doesn’t fit. CAPSTONE2 is the proof the composition operates end-to-end and fails safe.

19. CROSS-CUTTING THREADS (wired through EVERY stage – not per-stage afterthoughts)

- **Uncertainty thread (S2->S6):** CALIB1 (ordinal confidence) -> CONFORMAL1 (marginal is vacuous for the target class; even class-conditional degrades OOD) -> **RULE propagated to every stage: every output carries a confidence; on a novel organism/class, widen/cap it (COMPOSITE3 cap) and never claim nominal coverage OOD.**
- **Base-rate / study-bias thread:** FAIRGATE1 (risk-ratio gate) + META1 + the study-bias controls (F3CLIN1/GENETICS1/MET4) -> **RULE: every enrichment/relevance claim uses base-rate-fair + study-bias-controlled statistics** (this is why several positives survived and several tempting ones were killed).
- **Safety thread (gate before S4):** hard host-non-homology filter (FRONT1/E2E2) -> **RULE: no target reaches intervention design without passing it** (conservation-ranking-as-safety is CLOSED – most-conserved = host-toxic).
- **Provenance/reproducibility thread (all stages):** pre-registration + reproduce-x2 + first-class negatives + LEDGER -> **RULE: every output tagged with evidence-tier + reproduce-sha; a result that fails to reproduce or a confound found is retracted immediately** (V14/N1, host-embedded rule, EXPDESIGN1-G2, CONFORMAL1-marginal all retracted/down-tiered).

20. FAILURE-CASCADE map (if a node fails, what downstream is affected -> the systemic response)

- **S1 mechanism fails for a class** (non-metabolic; host-scavenger GEM) -> S2 falls back to conservation-breadth or **abstains** -> S3/S4 run only on what S2 emits; if S2 abstains, the **whole pipeline abstains** (never fabricates). *Terminal: honest “no trustworthy target from computation; needs experimental data.”*
- **S4 affinity is OPEN/GPU-gated** -> S5 cannot validate a *novel* molecule -> route to **repurposing** (if a drug exists) or **experimental screening**; never emit a fake potent lead.
- **S5 wet-lab unavailable** -> the loop (?17) cannot close -> the engine stays bounded to in-silico hypotheses + validated domains; **publish the validated core + negatives** (the value that does NOT need the loop closed).
- **S6 mis-fire risk** -> caught by CAPSTONE2 fail-safe + 93 unit tests + DARK1; a detected mis-fire -> **abstain + fix before any claim. Every cascade terminates in ABSTENTION or EXPERIMENT-ROUTING – never in fabrication.** This is the integrity invariant of the whole graph.

21. Phase-1->Phase-2 trigger, resource ROI, and per-stage KPIs (so progress is measurable, not vibes)

- **Phase-1-complete trigger (computational):** all reachable classes covered-or-abstaining ? both halves built-or-honestly- bounded ? integration proven (CAPSTONE2 ?) ? every wall characterized ? GPU/data-gated items specced. *Status: the un-gated CPU portion is essentially done; ~20-25% of the fullest vision because the gated/open majority remains.*
- **Phase-2 trigger (real-world):** a collaborator + a nominated target => run CRISPRIDESIGN1 (O7). This is the **ONLY** trigger that starts the closed loop (?17).

- **Acquisition ROI ordering (highest first): GPU** (unlocks the 14-weight intervention half) ~ **wet-lab bench** (unlocks real-world + the loop) » curated eukaryote GEMs / Pf screen (unlock specific bounded gaps) » **more CPU experiments** (~0 marginal – the frontier is mapped).
- **Per-stage KPIs (measurable):** S2 = odds/risk-ratio + analyst-blind pass-rate (now 4/7) ; S3 = precision@k + abstention- integrity (0 mis-fire – held) ; S4 = modality accuracy (0.814) + repurposing coverage (~1-7%) ; S5 = reproduce-x2 (held throughout) + [future] wet-lab confirm-rate ; S6 = fail-safe pass (CAPSTONE2 ?) + test suite (93/93).

22. THE LOGICAL SPINE (the single “why” chain – every link justified by evidence)

Vision = best intervention for any disease => needs mechanism->target->intervention->validation (S1-S5) => but label-free transfer is bounded by biological invariants (**transfer-condition principle**, META1) => so build per-class validated signals + compose + abstain (COMPOSITE/CAPSTONE) => the one signal that breaks the conservation ceiling is mechanistic FBA-essentiality, and it is **metabolic-scoped** (MET1-3; non-metabolic CLOSED x4 + ensemble (MULTISIG1)) => target-ID is therefore validated but *narrow*, strongest for pathogens (VAL-ESS/BLIND) => the intervention half needs zero-shot affinity, which is **OPEN/GPU- gated** (HIT2/AFFINITY1) => so no validated novel molecule is producible on CPU => real-world truth needs a **wet-lab** result to close the loop (?17), which is **experiment-gated** (CRISPRIDESIGN1 ready) => therefore the honest reachable end on CPU+open-data is a **validated, reproducible target-ID method + an honest negative map + a fail-safe composing engine** (~3/10 real contribution, ~20-25% of the computational vision) => and the needle-movers are exactly two acquisitions (a **GPU**, a **bench**) + **publication** – each named, none faked. *Every arrow above is a committed experiment or audit; this is the connected chain from the vision to the next concrete action, with no missing dot.*

PART IV – THE FULLEST-VISION ARCHITECTURE (the complete component tree, honestly status-tagged)

*Honest correction: the S0-S6 pipeline in Parts I-III is INTERCEPTA’s CURRENT THIN SLICE – a target-ID-centric, mostly-pathogen, mostly-metabolic spine. The FULLEST vision (“best therapeutic intervention for ANY disease, from minimal data, all the way to the clinic”) is a far larger platform. This Part enumerates the ENTIRE architecture it must contain – 7 layers, ~70 components – so nothing is missing, each tagged: **BUILT - PARTIAL - SEED(toy) - MISSING - OPEN(unsolved science) - GPU-gated - DATA-gated - EXPERIMENT-gated - CLINICAL-gated**. The current slice is a small highlighted subset; most of the vision is MISSING/gated, and this Part says exactly what and why.*

Honest recomputation of completeness against THIS fullest architecture: the ~20-25% figure was of the narrower “fullest *computational* platform buildable without experiments.” Against the fullest **end-to-end vision** below (which includes all disease classes, all scales, all modalities, preclinical, and clinical), INTERCEPTA is ~5-8% – deep in one sliver (Layer C metabolic target-ID + Layer F methodology), essentially empty in Layers A(breadth), D(most modalities), E(translation), and much of B and F. This is the honest denominator the earlier % understated.

Layer A – Disease universe & multimodal representation (accept ANY disease)

- **A1 Disease taxonomy across ALL classes:** infectious (bacterial [BUILT]/viral PARTIAL/fungal PARTIAL/parasitic PARTIAL/prion MISSING), cancer (PARTIAL: dependency), genetic (monogenic PARTIAL/complex PARTIAL), immune/autoimmune/inflammatory MISSING, metabolic-disease MISSING, cardiovascular MISSING, neurodegenerative/psychiatric MISSING, developmental/rare beyond-gene MISSING, aging MISSING, environmental/toxic MISSING, idiopathic/**never-seen** (the north-star case: PARTIAL via abstention only).
- **A2 Multimodal disease input:** genome [BUILT] - transcriptome SEED - proteome SEED - phenotype/HPO PARTIAL(PHENO1) - GWAS PARTIAL(GENETICS1) - CRISPR-screen PARTIAL(DEPEND1) - **epigenome** / **metabolome** / **microbiome** / **single-cell** / **spatial** / **imaging** / **EHR-clinical** / **epidemiology** / **patient-trajectory** = MISSING.
- **A3 Disease-state & context representation** (tissue, stage, comorbidity, patient sub-population): **MISSING**.

Layer B – Multi-scale biological understanding (mechanism at every scale)

- **B1 Molecular:** sequence [BUILT] - static structure PARTIAL(off-the-shelf AF/fpocket) - **dynamics/MD OPEN/GPU-gated** - **complexes/co-folding GPU-gated** - biophysics MISSING.
- **B2 Networks:** metabolism **BUILT** - transcriptional regulation CLOSED-as-target-signal(REGNET1) but as a MODEL MISSING - **signaling MISSING** - **epigenetic regulation MISSING** - PPI/interactome PARTIAL-negative(MET4) - **immune networks MISSING** - host-pathogen PARTIAL-negative(HOSTCTX).
- **B3 Cell: whole-cell models MISSING** - cell-state/single-cell MISSING - cell-cell communication MISSING.
- **B4 Tissue/organ/organism:** physiology/spatial MISSING - **PK/PD MISSING** - multi-organ MISSING.
- **B5 Population/evolution:** GWAS PARTIAL(GENETICS1) - **resistance/evolution dynamics PARTIAL** - DYNAMICS1 contact-residue durability signal (AUROC 0.84, n-fragile) works where AMR1 whole-protein static-biology fails) - epidemiology MISSING.
- **B6 Causal:** association PARTIAL(GENETICS1) - **causal graphs / counterfactuals / Mendelian-randomization / perturbation modeling = OPEN/MISSING** (a whole domain, ~untouched).

Layer C – Target / intervention-point identification (the current strength – but narrow)

- **C1 Essentiality:** metabolic **BUILT**; non-metabolic CLOSED x4 + ensemble (MULTISIG1) (conservation-breadth is the ceiling).
- **C2 Functional dependency:** **BUILT**; does not transfer label-free (TRANSFER1).
- **C3 Genetics/causal-gene:** PARTIAL(GENETICS1/MENDEL1). **C4 Structural class-ID:** PARTIAL(viruses).
- **C5 Systems/network targets** (master regulators, network controllability, **combination/multi-target**): MISSING/negative.
- **C6 Non-protein & hard targets** (RNA, DNA, PPI interfaces, **allosteric, degradation-tractable**): MISSING.
- **C7 Synthetic lethality / collateral vulnerability:** SEED(INTERVENE3/SYNLETH) - context-specific targets MISSING.

Layer D – Intervention design (the fullest MODALITY space – mostly MISSING)

- **D1 Small molecules:** de-novo generation SEED(B33/B40) - **novel-target binding affinity – GPU-gated even at small-candidate scale** (empirically confirmed: CPU pilot 0/20 in ~48 min, retracted the “CPU-feasible” reframe; HIT2/AFFINITY1/CPU_PILOT_FINDING) - ADMET [BUILT]-standalone - synthesizability [BUILT]-standalone - developability PARTIAL.
- **D2 Repurposing:** PARTIAL-narrow(INTERVENE1/2). **D3 Modality selection:** PARTIAL(MODALITY1, triage-only).
- **D4 Biologics** (antibodies/proteins): MISSING. **D5 Nucleic-acid** (ASO/siRNA/mRNA/gene-therapy/**gene-editing**): MISSING (only named in MODALITY1’s taxonomy). **D6 Targeted degradation** (PROTAC/molecular-glue): MISSING. **D7 Peptides/macrocycles:** MISSING. **D8 Cell therapies** (CAR-T): MISSING. **D9 Combinations:** SEED(synergy B24-29). **D10 Delivery/formulation:** MISSING.

Layer E – Validation & translation (in-silico -> clinic -> real-world)

- **E1 In-silico validation:** **BUILT**. **E2 Experiment design/active-learning:** PARTIAL(EXPDESIGN1), loop bounded(SIL2). **E3 Wet-lab:** EXPERIMENT-gated (CRISPRIDESIGN1 turnkey design ready). **E4 Preclinical** (organoid/ animal/tox): MISSING/EXPERIMENT-gated. **E5 Clinical-trial design & simulation, biomarkers, patient stratification:** MISSING/CLINICAL-gated (response-prediction tested-NEGATIVE, B10/B20). **E6 Regulatory / real-world-evidence / pharmacovigilance:** MISSING.

Layer F – Meta / AI capabilities (cross-cutting engines)

- **F1 Foundation models** (sequence/structure/chem/multimodal): SEED(ESM embeddings used; PLMESS1) - training/large models GPU-gated. **F2 Reasoning / agentic orchestration:** MISSING. **F3 Causal inference engine:** OPEN/MISSING.
- **F4 Simulation** (MD, systems-dynamics, PK/PD, evolutionary): OPEN/GPU-gated. **F5 Active/continual/closed-loop learning:** SEED(SIL1/2) bounded to in-domain. **F6 Uncertainty/calibration:** PARTIAL**BUILT**. **F7 Explainability/provenance:** [BUILT]. **F8 Multi-objective optimization:** SEED(BESTINT1). **F9 Knowledge integration/graphs:** PARTIAL. **F10 Reproducibility/governance:** **BUILT**. **F11 Safety/ethics/bias governance:** PARTIAL(host-safety filter; abstention) - broader ethics MISSING.

Layer G – Deployment & real-world operation

- **G1 Any-disease routing + abstention:** PARTIAL**BUILT**. **G2 Clinician/researcher interface:** MISSING(CLI only). **G3 Continuous update from new data (the closed loop):** SEED, EXPERIMENT-gated (?17). **G4 Collaboration/ data-sharing infrastructure:** PARTIAL(docs). **G5 Equity/access/cost, global-health deployment:** MISSING.

How Part IV connects back to the S0-S6 slice (the honest mapping)

The S0-S6 pipeline = {A1-partial, A2-partial} -> {B2-metabolism, B6-association} -> {C1-C4} -> {C-safety + F6/F8} -> {D1-affinity-gated, D2/D3} -> {E1, E2-seed, E3-gated} -> {F/G governance}. **Everything in Layers A(breadth), B(scales beyond metabolism), C5-C7, D4-D10, E4-E6, F1-F5(most), G2/G5 is MISSING or gated** – that is the “lots and lots missing” you named, now enumerated so it is not a vague gap but a concrete, prioritized backlog.

What this reveals for “what to do / what’s next” (honest, non-random)

1. **The vision is ~5-8% built end-to-end** – most of it is MISSING (whole modalities, scales, disease classes, translation) or OPEN/gated (affinity, causal, simulation, foundation-model training, wet-lab, clinical). The earlier ~20-25% was of a narrower denominator; against the fullest architecture it is smaller, and I state that plainly.
2. **The MISSING items split the same 3 ways (?11):** most are **ACQUISITION** (data modalities A2, GPU for D1/F1/F4, wet-lab/clinical for E3-E6) or **genuine OPEN inventions** (causal B6/F3, novel-target affinity D1, dynamics F4) – only a thin engineering layer is pure CPU-buildable now (e.g., broaden A1 disease-class arms where open data exists; F9 knowledge graph; F5 loop hardening). **This confirms: more CPU experiments cannot build the fullest vision – data, GPU, wet-lab, and a few field-level open inventions are the real content.**
3. **Highest-leverage additions to the CURRENT slice, in order (each honestly tiered):** (a) GPU -> D1 affinity + F1/F4 (the biggest computational chunk); (b) wet-lab -> E3 -> closes the loop (?17); (c) open-data CPU arms for uncovered A1 classes where a validated signal plausibly exists (immune/metabolic-disease via genetics – extends GENETICS1) – modest but genuine;
(d) causal engine (F3/B6) – a genuine open-science build, attempt only with a study-bias-robust design.
4. **What we will NOT pretend:** we will not claim Layers D4-D10 / E4-E6 / B3-B4 exist or are near – they are MISSING and most need biology, data, or experiments we do not have. Naming them completes the map; building them is gated, not imminent.